INCH-POUND

MIL-DTL-60315C (AR)
2 October 2007
SUPERSEDING
MIL-C-60315B (AR)
9 September 1991

DETAIL SPECIFICATION

CHARGE, PROPELLING, M67, FOR 105MM HOWITZER LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Armament Research, Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for one type of propelling charge designated as Charge, Propelling, M67, for Howitzer, M2A2, M103 and M137.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to (Commander ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, NJ 07806-5000) or emailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk**,** 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

9205470	-	Bag
9205471	-	Increment Loading Assembly
9205472	-	Charge, Propelling, M67 Assembly
9288695	-	Charge, Propelling, M67 Assembly
9288696	-	Increment Loading Assembly
9288697	-	Bag
9345265	-	Container, Steel
9381476	-	Packing and Marking for Containers, Steel for Bulk Propellants

(Copies of these drawings are available from US Army ARDEC, AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000, or email Drawing-Request@pica.army.mil)

INTERNATIONAL TEST OPERATING PROCEDURES

ITOP-4-2-504 (1) - FR/GE/UK/US Safety Testing of Field Artillery Ammunition

(Copies of these documents may be ordered from the US Army Developmental Test Command, Attn: Publications, 314 Longs Corner Road, Aberdeen Proving Ground, MD 21005-5005 or online at http://www.dtc.army.mil/publications/topsindex.aspx .)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supercedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First Article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Propelling Charge Assembly. The propelling charge assembly shall comply with requirements specified on Drawings (Dwg.) 9345265, 9381476, and 9205472 or 9288695.

3.3 Increment loading assembly weight. The net weight of the propellant shall not differ by more than the amounts specified in Table I from the loading authorization:

Table I. Weight tolerances for propelling charge increments

Increment Number	Tolerance (ounces)
1	0.02
2	0.015
3	0.015
4	0.015
5	0.015
6	0.02
7	0.02

3.4 Charge, propelling assembly weight. The charge, propelling assembly weight shall not differ by more than 0.10 ounces from the total of the amount specified in the loading authorization, plus tare weight of bags and cord. (See 6.7)

3.5 Workmanship. All parts and assemblies (including packaging) shall be fabricated and finished in a thorough, workmanlike manner. All parts shall be free of dirt, grease, and other foreign material. All required markings shall be neat and sharply defined.

3.5.1 Container pressure seal. Gaskets for container lids shall not be cut, dried-out, cracked, abraded or improperly affixed or affixed with excessive adhesive. Containers shall be sealed from external moisture. Sealed containers shall withstand an internal pressure of 2.0 - 0.5 pound per square inch (psi) without a pressure drop of ¼ psi or more (0.7 inch or more on the manometer) for a period of 15 seconds.

3.6 Proving ground. The muzzle velocity, pressure, and velocity standard deviation shall be as specified in Table II:

Table II. M67 Velocity and pressure in 105MM Howitzers, M101, M102 and M119

Zone	Temp (°F)	Velocity (fps), (corrected)			Individual Pressure (psi), (corrected)		Average pressure (psi), (corrected)	
		Min	Max	Stan.Dev.	Min	Max	Min	Max
7	-50	NA	NA	NA	NA	44,850	NA	NA
7	+70	1500	1550	8	NA	44,850	31,200	39,000
7	+145	NA	NA	NA	NA	44,850	NA	NA
NA – Not applicable.								

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4. VERIFICATION

Table III. Requirements/verification cross reference matrix

Method of Verification	Classes of Verification
1- Analysis 2- Demonstration (end item test) 3- Examination 4- Test	A - First Article B - Conformance

Section 3 Requirements	Description	Verification Method				Verification Class		Section 4
		1	2	3	4	A	B	
3.1	First Article		X	X	X	X		4.2
3.2	Propelling charge assembly			X	X	X	X	4.3
3.3	Increment loading assembly weight			X		X	X	4.3
3.4	Charge, propelling assembly weight			X	X	X	X	4.3
3.5	Workmanship			X	X	X	X	4.3
3.6	Proving ground		X	X	X	X	X	4.4.4

4.1 Classification of inspection. The verification requirements specified herein are classified as follows:

a. First Article Inspection (see 4.2)

b. Conformance Inspection (see 4.3)

4.2 First Article. A sample shall be subjected to first article verification in accordance with 4.2.1

4.2.1 Inspections to be performed. When specified, the first article samples will be subjected to any or all of the examinations and tests specified in the specification and inspected for compliance with all of the requirements of this specification. Inspections to be performed are specified in 4.3.2.1, 4.3.2.2, 4.3.2.3, 4.3.2.4, 4.3.2.5, 4.3.2.6, and 6.8 with Table IV.

Table IV First article test

Test Phase	Temp (°F)	No. Control Rounds	No. Test Rounds	TECOM TOP
Propellant Checkout	70		10	ITOP-4-2-504(1)
	145		10	ITOP-4-2-504(1)
	-50		10	ITOP-4-2-504(1)
Sequential Environmental	145	10	20	ITOP-4-2-504(1)
	-50	10	20	ITOP-4-2-504(1)

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4.2.2.1 Rejection. The first article sample shall be rejected if the sample fails to meet any of the requirements.

4.3 Conformance Inspection.

4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-DTL-1916 specification. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded charge assemblies shall contain:

a. Single perforated propellant from not more than one lot.

b. Multi perforated propellant from not more than one lot. An entire lot of multi-perforated propellant shall be utilized in the formation of one lot of propelling charges unless otherwise specified.

4.3.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Inspection Level IV shall be used for all characteristics defined as Majors and Inspection Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with Paragraphs 4.3.2.1 to 4.3.2.6.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.1	TITLE Bag increment	SHEET 1 OF 1		DRAWING NUMBER 9205470 NEXT HIGHER ASSEMBLY 9205471
CATEGORY	EXAMINATION OR TEST	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major				
101	Foil missing or improperly assembled (applicable to increment 5 only).	IV	3.2	Visual
Minor				
201	Any seam or opening incompletely stitched	II	3.2	Visual
202	Marking misleading or unidentifiable	II	3.2	Visual
203	Evidence of poor workmanship	II	3.5	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.2	TITLE Bag increment	SHEET 1 OF 1		DRAWING NUMBER 9288697 NEXT HIGHER ASSEMBLY 9288696
CATEGORY	EXAMINATION OR TEST	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major	None defined			
Minor				
201	Any seam or opening incompletely stitched	II	3.2	Visual
202	Marking misleading or unidentifiable	II	3.2	Visual
203	Evidence of poor workmanship	II	3.5	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.3	TITLE Increment loading assembly	SHEET 1 OF 1		DRAWING NUMBER 9205471 or 9288696 NEXT HIGHER ASSEMBLY 9205472 or 9288695
CATEGORY	EXAMINATION OR TEST	VERIFICATION LEVEL	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None Defined			
Major				
101	Assembly damaged to the extent that propellant can escape	IV	3.2	Visual
Minor				
201	Any seam broken or incompletely stitched	II	3.2	Visual
202	Thread broken	II	3.2	Visual
203	Evidence of poor workmanship	II	3.5	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.4	TITLE Charge, Assembly	SHEET 1 OF 1		DRAWING NUMBER 9205472 or 9288695 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	VERIFICATION LEVEL or %	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical				
1	Missing or incorrect increments or incorrect propelling charge assembly weight	100% (see Note 1)	3.4	Scale (see 4.4.1)
Major				
101	Incorrect sequence of increments	100%	3.2	Visual
102	Assembly damaged to the extent that propellant can escape	100%	3.2	Visual
Minor				
201	Increments insecurely tied at junctions	II	3.2	Visual
202	Ties loose, broken or insecurely tied	II	3.2	Visual
203	Marking missing or unidentifiable	II	3.2	Visual
204	Evidence of poor workmanship	II	3.5	Visual

1/ The defect shall be classified critical when the excess in weight is more than 3 times the allowable tolerance as specified in 3.4. (See 6.9). Major when the weight is in excess of that in 3.4 but not greater than 3 times. Compliance with 3.4 shall be determined by using the average weight of the empty bag assembly. If the charges are assembled in accordance with 9205472 and the defect is determined to be due to an incorrect propelling charge assembly weight, remove increment 5 by raising it off the scale without breaking the cord and replace it with a weight representing the average weight of increment 5. If the weight is incorrect, reject the charge. If the weight is correct and it can be determined that the off weight of increment 5 is caused by the lead foil (tare weight of bag) the charge shall be considered acceptable.

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Conformance inspection by Classification of Characteristics

PARAGRAPH 4.3.2.5	TITLE Drum, prior to filling	SHEET 1 OF 1		DRAWING NUMBER 9345265 NEXT HIGHER ASSEMBLY 9381476
CATEGORY	EXAMINATION OR TEST	VERIFICATION LEVEL or %	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major				
101	Foreign material in drum	100%	3.2	Visual
102	Inspection of gasket	100%	3.2 & 3.5	Visual, (see 4.4.2)
103	Pressure test of container	100%	3.5	4.4.3
Minor	None defined			

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.6	TITLE Drum sealed	SHEET 1 OF 1		DRAWING NUMBER 9345265 and 9381476 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	VERIFICATION LEVEL or %	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical	None defined			
Major				
101	Locking ring improperly positioned	100%	3.2	Visual
Minor				
201	Seal missing or improperly positioned	II	3.2	Visual
202	Marking misleading or unidentifiable	II	3.2	Visual

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4.4 Methods of inspection.

4.4.1 Increment loading assembly weight. The increment shall be weighed and check weighed 100 percent. Any increment which fails to comply with the requirements shall be classed defective and removed from the lot. Inspection equipment will be in accordance with 4.4.5.

4.4.1.1 Method 1. The propellant shall be weighed in a container and placed in the bag, the bag sewn and the increment loading assembly weighed. The net weight of propellant may be calculated using an average tare weight of the empty bags or the actual tare weight as marked on the individual bag.

4.4.1.2 Method 2. The propellant in a container shall be weighed and then check weighed on a different balance and shall then be placed in the bag and the bag sewn. The weight of the loaded bags should be checked as a major defect in accordance with MIL-STD-1916. The net weight of the propellant shall be calculated using an average weight of the empty bag or the actual tare weight as marked on the individual bag.

4.4.2 Container gasket inspection. Container lids will be inspected 100% for cut, dried-out, cracked, abraded or improperly affixed gaskets or excessive adhesive used in affixing gaskets. If any gasket is found with any of the conditions listed above, the lid shall be classed defective and removed from the lot.

4.4.3 Container air pressure test. All containers (100%) shall be subjected to an internal pressure test of 2.0 - 0.5 pounds per square inch (psi) by an approved method, (see 6.4). A gage (designed to read ¼ psi) or a water manometer shall be assembled in the system. A container experiencing a drop of ¼ psi on the gage or 0.7 inches on the manometer in 15 seconds shall be classed defective and removed from the lot.

4.4.4 Proving ground assessments. These tests shall be conducted at a government proving ground in accordance with the appropriate acceptance test procedure (ATP) using inert M1 projectile (33.00 +/- 0.02 pounds) and fired from an M101A1 Howitzer with a M2A2 Cannon with 75 percent minimum remaining life. Prior to firing, all charges shall be temperature conditioned for a minimum of 24 hours and shall be fired within five minutes after removal from the conditioning chamber. A tolerance of -5 Fahrenheit degrees applies to all -50°F and +145°F requirements and a tolerance of plus or minus 2 Fahrenheit degrees applies to all 70°F requirements. Calibration charges shall be fired alternatively with the test samples and the muzzle velocity and pressure of all charges shall be corrected to standard condition.

4.4.4.1 Muzzle velocity and muzzle velocity standard deviation. If the average velocity at 70 degrees F is below the minimum velocity of 1500 fps (457.2 m/sec) or exceeds the maximum velocity of 1550 fps (472.4 m/sec) at Zone 7, the lot shall be rejected. If the velocity standard deviation of the samples exceeds 8 fps (2.4 m/sec), the lot shall be rejected.

4.4.4.2 Chamber pressure.

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4.4.4.2.1 Permissible individual maximum pressure. If any individual maximum pressure at any operating temperature (-50°F to +145°F) exceeds 44,850 psi, the lot shall be rejected.

4.4.4.2.2 Averaged corrected pressure. If any propelling charge average corrected pressure obtained at +70 degrees is below the minimum pressure of 31,200 psi or exceeds 39,000 psi, the lot shall be referred to U.S. Army Picatinny.

4.4.5 Inspection equipment. Inspection equipment required to perform the inspections prescribed in this specification is identified in the Examinations and Test Method paragraphs herein. See 6.6 for details concerning responsibilities for inspection design and approval.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2f). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The M67 propelling charge covered in this specification is intended for use in 105MM ammunition for Howitzer, M2A2, M103 and M137.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

c. Requirements for Ammunition Lot Numbering, if applicable.

d. Requirement for acceptance inspection equipment (AIE) designs.

e. Certificate of Conformance for each lot or shipment of product.

f. Preservation, packaging, and marking. See 6.3 and ARDEC Drawing 9381476 – “Packing and Marking for Containers, Steel for Bulk Propellants.” and ARDEC Drawing 9345265 - "Container, Steel"

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6.3 Acceptable packing and marking requirements. The following packaging has been used, and is found to be acceptable. (Caution - When used as requirements, the paragraph should be revised incorporating contractually compulsory languages.).

6.3.1 Level A. The propelling charges should be packed in containers conforming to MS27683-2. The gross weight of the pack should not exceed 150 pounds. Cushioning material, conforming to A-A-1898B, Cushioning Material, Cellulose, Packaging, should be added as required to provide a tight pack.

6.3.1.1 Reworked containers. In addition to meeting the requirements of 9345265, the gasket of all reworked containers should be replaced in accordance with 9345265 prior to reuse.

6.3.2 Air pressure test of containers. Immediately after packing, all containers should be tested with compressed air at an internal air pressure of 2.0 – 0.5 psi. The container should withstand the specified pressure for a minimum of 15 seconds without apparent leakage as defined in 4.4.3.

6.3.3 Marking of steel drums. Containers should be marked in accordance with 9381476. The Proper Shipping Name and Identification Number (PSN/IN) should be "CHARGES, PROPELLING, FOR CANNON UN0242."

6.4 Submission of inspection equipment designs for approval. Submit equipment designs as required per the Contract Data Requirements Lists (DD Form 1423).

6.5 Submission of ballistic test report and ammunition data cards. A proving ground ballistic test report and ammunition data cards should be submitted to:

Commander, ARDEC
ATTN: AMSRD-AAR-AEE-W, Propulsion Research and Engineering Branch
AMSRD-AAR-QEM-C, Quality Engineering and System Assurance
Picatinny, NJ 07806-5000.

6.6 Inspection Equipment Designs. Inspection equipment designs are of two types – Government Special Inspection Equipment (SIE) and contractor designs. The SIE designs, if any, are designated by drawing numbers under the "Test Methods and Procedures" heading in section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.6.2 and 6.6.3 should he elect to do so.

6.6.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawing which completely depicts all information necessary for the fabrication and use of the item of inspection equipment.

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b. A source control drawing or a specification control drawing as defined in ASME Y14.100.

c. An envelope drawing, as defined in ASME Y14.100, which establishes the criteria which a detailed design should meet. When envelope drawings are specified, the contractor should prepare designs which comply with the criteria therein.

6.6.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposed to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs should include appropriate operating instructions, calibration procedures, maintenance procedures, and Reliability & Repeatability Studies consisting of data and methodology used. Commercial equipment should be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs should be prepared for any special fixtures required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.5.1c). Designs should be of the category and form (per DOD-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, “Inspection Equipment Design” for guidance.

6.6.3 Submission of Designs for Government approval. Contractor designs should be approved by the Government prior to fabricating or procuring the equipment. Designs should be submitted for approval in accordance with the Contract Data Requirements Lists (DD Form 1423) or in the contract. The contractor must include in his submission of designs a listing of any and all Government SIE designs which will be used. Partial submission of inspection designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs.

6.7 Charge, propelling assembly weight. This requirement applies to the load plant only due to change in tare weight of the bags because of variation in bag material and moisture content.

6.8 First Article Inspection. The first article test is a one time qualification test for a new contractor to load, assemble and pack acceptable M67 propelling charges. Once a contractor has demonstrated the ability to load, assemble and pack the propelling charges, the first article test may be waived by the contracting officer. Test samples should be withdrawn from the first lot of propelling charges produced by the contractor. Charges will be tested in accordance with all of the requirements of this specification and with Table IV. The test matrix attached as Figure 1 should be used for the sequential environmental phase of the test. Testing will be conducted in accordance with TECOM ITOP-4-2-504. Until the first article samples have been found to be acceptable, the contractor is in no way authorized by the Government to initiate production of a second lot unless otherwise directed by the contracting officer.

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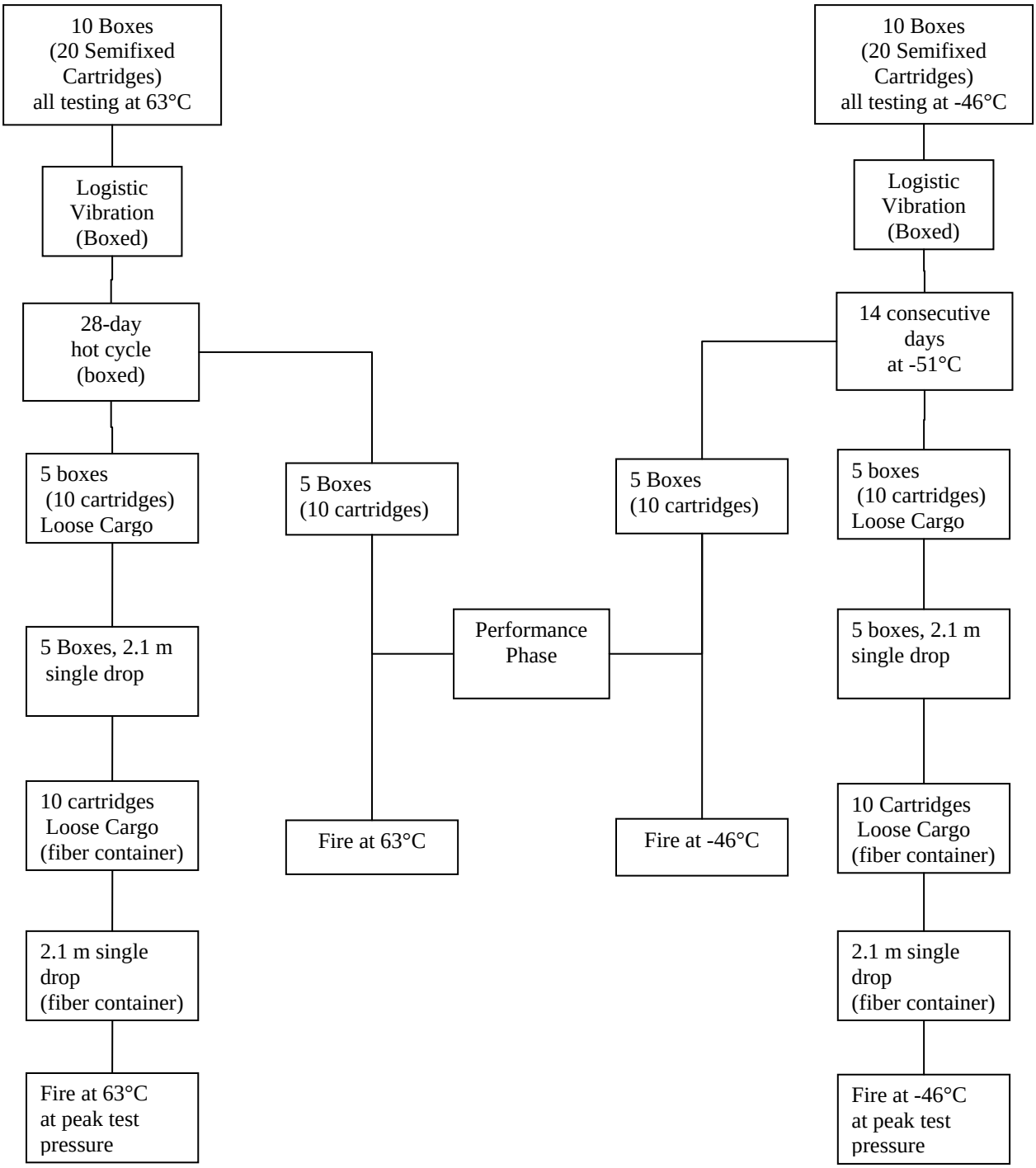


Figure 1. Sequential test for first article of M67 Propelling Charge

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6.9 Critical characteristics. Excess in propellant weight could cause temperature and pressure of the gun tube to escalate. An overpressure gun tube at a higher than normal operating temperature could cause gun tube deformation, and is hazardous to the energetic warhead. This characteristic (excess in propellant), if out of control, could cause hazardous and unsafe condition for the gun crew especially during sustained rapid firings at a hot climatic condition. Missing increment or incorrect increment or incorrect weight of the propelling charge which results in a lower weight for the propelling charge could cause short rounds which could endanger the gun crew or cause a friendly fire incidence.

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.11 Keyword list.

Propellant charge
M1 Propellant

Custodian:
Army-AR

Preparing Activity:
Army-AR
(Project 1315-2007-005)

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